

Jawaharlal Nehru Govt. Engineering College Sundernagar, Mandi (H.P.)

Course Name: Computer Networks Course Code: CSPC-511 Examination: 1st Mid Semester Test

Scheme: NEP Branch: B.Tech CSE Semester: 5th

Question Paper for 1st Mid Semester Test (October, 2025)

Time: One and half hour

Max. Marks: 30

Date: 01.10.2025

Note: Attempt all questions.

Course Outcomes:

CO1. Understand the basic network technology and functions of each layer in the OSI model and TCP/IP reference model.

CO2. Learn and apply the basic concept of data link layer.

CO3. Compare various sliding window protocols.

CO4. Implement the addressing techniques and network routing protocols.

CO5. Apply the concept of transport layer and compare the transport protocols.

CO6. Familiarities with the essential protocols of session, presentation and application layer.

Q.1 Answer the following:

(2x5=10)

- (a) Define LAN and WAN with one example each?
- (b) What is the function of multiplexing in data communication?
- (c) Write 2 functions of Data Link Layer?
- (d) What is the significance of flow control?
- (e) List the elements of protocols?

Q.2 Explain ISO-OSI model with neat and clean diagram.

(5)

Q.3 Differentiate between Guided and Unguided Media.

(5)

Q.4 If frame = 10, window size = 3, every 4th packet is lost. Find the total number of messages sent from sender to receiver by using Go-Back-N technique.

(5)

Q.5 Explain Go-Back-N and Selective Repeat Protocol with neat diagram.

(5)